

AUBURN UNIVERSITY, AL, USA

WMO: 722284

Lat: 32.616N

Lon: 85.433W

Elev: 237

StdP: 98.52

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03892

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.9	-2.5	-14.7	1.1	-0.5	-12.3	1.3	0.8	9.2	8.2	8.4	7.3	3.8	340	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	34.1	23.2	32.9	23.4	32.3	23.3	25.5	31.1	25.0	30.5	24.5	29.6	3.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.3	27.6	23.1	18.4	26.9	22.8	18.1	26.6	78.9	31.6	76.9	30.3	75.2	29.2	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	7.1	6.0	DB -9.0	35.5	2.5	2.2	-10.8	37.1	-12.2	38.3	-13.6	39.6	-15.5	41.2		
(5)			WB -10.0	26.1	2.3	0.7	-11.7	26.6	-13.0	27.0	-14.3	27.4	-16.0	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	7.9	10.1	13.5	17.4	21.9	25.2	26.6	26.3	23.7	18.4	12.9	9.2
	DBStd	7.74	5.67	5.21	4.77	3.78	3.16	2.22	1.86	2.03	2.80	4.02	4.49	5.05
	HDD10.0	293	109	59	23	2	0	0	0	0	0	1	22	77
	HDD18.3	1290	323	233	159	61	9	0	0	0	2	49	168	284
	CDD10.0	3144	44	61	132	223	370	457	516	506	410	263	109	53
	CDD18.3	1099	1	2	9	32	121	207	257	248	162	53	6	2
	CDH23.3	9116	1	6	59	237	964	1789	2352	2173	1192	319	23	2
	CDH26.7	3493	0	0	6	29	297	737	1036	920	411	56	2	0
(14) Wind	WSAvg	2.9	3.3	3.4	3.4	3.0	2.7	2.4	2.4	2.3	2.6	2.8	2.9	3.2
(15) Precipitation	PrecAvg	1317	120	119	158	102	95	109	128	103	90	79	102	110
	PrecMax	1927	219	196	359	185	244	258	279	242	220	267	259	302
	PrecMin	949	60	42	25	47	8	39	56	43	16	5	36	46
	PrecStd	235	43	39	73	42	59	60	58	46	47	64	52	53
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.6	24.1	27.4	28.9	32.7	35.2	36.2	37.0	33.8	29.9	26.2	22.9
		MCWB	17.1	17.1	17.6	19.7	21.8	22.5	23.5	23.9	21.7	21.7	18.5	17.6
	2%	DB	20.1	22.1	25.0	27.5	31.2	33.4	34.1	34.0	32.4	27.9	23.6	21.0
		MCWB	15.7	16.3	16.7	18.6	21.5	22.9	23.6	23.8	22.1	20.1	16.8	17.2
	5%	DB	18.0	19.9	22.9	26.2	29.7	32.4	32.8	32.6	31.0	27.1	22.2	18.9
		MCWB	14.8	15.2	15.3	17.8	21.0	23.0	23.9	23.9	22.2	19.4	16.2	16.4
	10%	DB	17.0	17.9	21.4	24.3	28.0	31.1	32.1	31.3	28.9	25.1	20.2	17.4
		MCWB	13.9	13.8	14.7	16.9	20.2	22.8	23.8	23.6	21.9	18.6	16.0	14.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.4	19.5	19.7	21.8	23.9	25.7	26.1	26.2	24.7	23.7	21.2	19.8
		MCDB	20.3	22.1	23.8	25.9	28.3	31.0	31.4	32.0	29.6	27.5	22.4	20.9
	2%	WB	17.2	18.0	18.6	20.4	22.9	25.0	25.4	25.5	24.1	22.5	19.5	18.3
		MCDB	19.0	20.2	22.1	24.7	28.1	30.4	31.2	31.2	28.5	25.6	21.6	19.6
	5%	WB	15.9	16.7	17.4	19.5	22.2	24.3	25.0	24.9	23.5	21.4	18.0	16.9
		MCDB	17.8	18.9	21.2	23.9	27.3	29.5	30.6	30.4	27.7	24.3	20.3	18.3
	10%	WB	14.5	15.2	16.1	18.6	21.6	23.7	24.5	24.4	22.9	20.3	16.5	15.3
		MCDB	16.5	17.8	19.9	22.8	26.4	28.6	29.7	29.4	26.8	23.3	19.2	17.5
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	10.5	11.3	11.5	10.8	9.9	9.8	9.5	9.7	10.7	10.9	9.7
		MCDBR	10.3	11.1	12.7	12.5	12.0	11.7	11.3	11.3	11.5	11.6	11.7	10.0
		MCWBR	7.2	7.0	6.7	5.8	4.5	4.1	3.6	3.7	4.2	5.2	6.7	7.2
	5% WB	MCDBR	8.7	9.4	10.3	10.0	9.9	9.8	9.9	9.7	8.8	8.5	9.1	8.7
		MCWBR	7.2	7.4	6.7	5.6	4.5	4.2	3.7	3.7	4.0	5.0	6.7	7.4
(40) Clear-Sky Solar Irradiance	taub		0.317	0.320	0.349	0.380	0.433	0.473	0.512	0.512	0.417	0.365	0.341	0.321
	taud		2.531	2.520	2.455	2.354	2.248	2.189	2.091	2.083	2.337	2.464	2.476	2.531
	Ebn at Noon		907	938	930	909	860	821	787	781	849	875	867	880
	Edh at Noon		84	93	107	124	139	147	162	161	120	98	88	80
(44) All-Sky Solar Radiation	RadAvg		2.80	3.36	4.49	5.63	6.21	6.14	6.03	5.50	4.84	4.13	3.14	2.39
	RadStd		0.17	0.40	0.38	0.39	0.48	0.48	0.41	0.32	0.42	0.53	0.29	0.26

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+156	+106

Nomenclature: See separate page